County geography adjustments inside the CH index

Controls: CH specification—County-adjusted CH; Geography; County calibration instrument; Include log county area in the calibration.

1. Estimate geography effects using counties. This option estimates geography effects where there are many county observations, then incorporates those effects into the MSA or state index. For county c, let q_c be all-industry GDP, n_c total employment, a_c land area and $d_c = n_c/a_c$ density. The linked county regression is

$$\log(q_c/n_c) = \alpha_{s(c)} + \rho_C \log d_c + X_c'\gamma + \lambda \log a_c + u_c.$$

Here $\alpha_{s(c)}$ is a modern state fixed effect, X_c contains the selected geography and transport variables, γ their coefficients, and u_c the error. Log county area is optional and off by default. Its coefficient is λ. The county density coefficient is ρ_C.

2. The calibration instrument and sample. With IV, historical county population in levels instruments log density. The population construction can be selected separately from the regional instrument. The county regression uses the same historical sample year, instrument year and territory scope as the regional model. With OLS, it estimates the same county equation without an instrument.

For State analysis, calibration uses eligible counties assigned to the retained historical regions. For MSA analysis, it uses eligible counties in the represented modern states, including nonmetropolitan counties. County slopes agree with standalone County IV when inputs, samples, controls and fixed effects agree. Geography variables are treated as exogenous controls.

3. Transfer geography effects into the index. For region g, total employment is $N_g = \sum_{c \in g} n_c$. Construct

$$D_g^{\text{geo}}(\theta) = \frac{1}{N_g} \sum_{c \in g} \exp(X_c'\widehat{\gamma})\, n_c^{\theta} a_c^{1-\theta}.$$

Only the estimated geography and transport effects enter the multiplier. County fixed effects, the optional area coefficient and ρ_C are not transferred. The regional regression relates log GDP per employment to $\log D_g^{\text{geo}}(\theta)$ and estimates its own elasticity $\rho = \theta - 1$. In MSA analyses, schooling and college can still adjust the regional outcome separately.

4. Missing inputs and uncertainty. Every constituent county remains in the index, even if its historical instrument is unavailable for calibration. A missing required input in a positive-employment county excludes the region. Joint errors account for geography estimation and shared county–regional observations. In State analysis, each historical region and its counties share a covariance group; modern state identities remain county fixed effects. The county calibration must be identified. Transferring geography effects still requires common county coefficients.